

JENKINS-01

1. Install Jenkins and run Jenkins on port number 8081.

- First lets create an EC2 instance

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name
Jenkins [Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents **Quick Start**

▼ Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....[read more](#)
ami-0a59ec92177ec3fad

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

[Cancel](#) [Launch instance](#) [Preview code](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0a59ec92177ec3fad (64-bit (x86), uefi-preferred) / ami-0345e0acbdfe50a10 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs [Free tier eligible](#)

▼ Summary

Number of instances [Info](#)
1

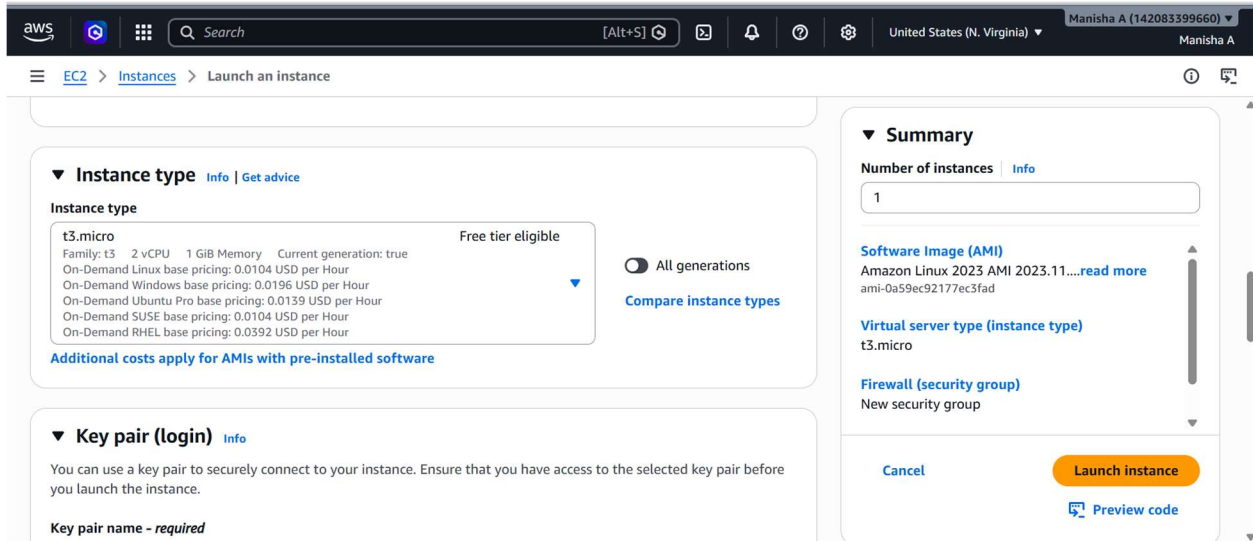
Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....[read more](#)
ami-0a59ec92177ec3fad

Virtual server type (instance type)
t3.micro

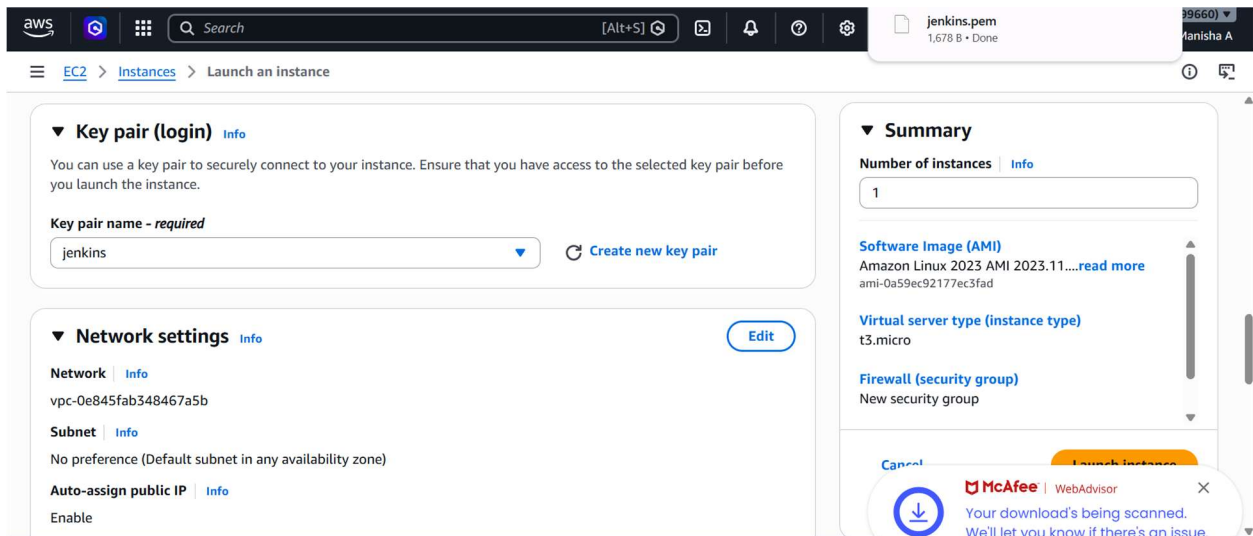
Firewall (security group)
New security group

[Cancel](#) [Launch instance](#) [Preview code](#)

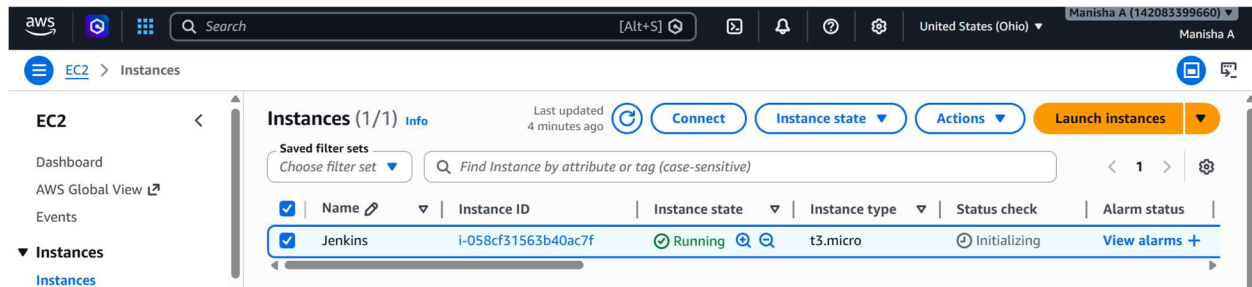
Chosed instance type t3.micro



Created a new key pair → default vpc → default security group



Gave 20 gb in configuration storage



```
root@ip-172-31-39-19:/home/ec2-user
```

```
manis@MANISHAREDDY MINGW64 ~/Downloads
m ssh -i "Jenkins-1.pem" ec2-user@ec2-52-14-13-20.us-east-2.compute.amazonaws.com
The authenticity of host 'ec2-52-14-13-20.us-east-2.compute.amazonaws.com (52.14.13.20)' can't be established.
ED25519 key fingerprint is SHA256:6ntnqwuVfIoXv9de71P7mQondWXTONb+zcVG4jfuwL4.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-52-14-13-20.us-east-2.compute.amazonaws.com' (ED25519) to the list of known hosts.
```

```
~\_#_
~\_####_ Amazon Linux 2023
~\_#####\
~\_###|
~\_#/\
~\_V~'_'>
~\_
~\_.-./
~\_/m/'
```

```
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-39-19 ~]$ sudo su
```

Add the Jenkins repo using the following command: `sudo wget -O /etc/yum.repos.d/jenkins.repo \`

<https://pkg.jenkins.io/rpm-stable/jenkins.repo>

```
root@ip-172-31-39-19:/home/ec2-user
[root@ip-172-31-39-19 ec2-user]# sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/rpm-stable/jenkins.repo
--2026-05-11 09:46:53-- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.78.133, 2a04:4e42:83::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)[146.75.78.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenki 100%[=====>]      267  --.-KB/s    in 0s
2026-05-11 09:46:53 (12.2 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]

[root@ip-172-31-39-19 ec2-user]# |
```

Importa key file from Jenkins-CI to enable installation from the package

```
[root@ip-172-31-39-19 ec2-user]# sudo rpm --import https://pkg.jenkins.io/rpm-stable/jenkins
.io-2026.key
[root@ip-172-31-39-19 ec2-user]# sudo yum upgrade
Jenkins-stable          6.0 kB/s | 833 B      00:00
Jenkins-stable          44 kB/s | 1.6 kB      00:00
Importing GPG key 0x14ABFC68:
  Userid      : "Jenkins Project <jenkinsci-board@googlegroups.com>"
  Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
  From        : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
Is this ok [y/N]: y
Jenkins-stable          13 kB/s | 1.9 kB      00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-39-19 ec2-user]# |
```

Install Java


```
root@ip-172-31-39-19:/home/ec2-user

[root@ip-172-31-39-19 ec2-user]# sudo yum install java-21-amazon-corretto -y
Last metadata expiration check: 0:01:05 ago on Mon May 11 09:48:30 2026.
Dependencies resolved.
=====
Package                                Arch      Version                                Repository    Size
=====
Installing:
java-21-amazon-corretto                x86_64    1:21.0.11+10-1.amzn2023.1            amazonlinux   218 k
Installing dependencies:
alsa-lib                               x86_64    1.2.7.2-1.amzn2023.0.3                amazonlinux   503 k
cairo                                  x86_64    1.18.0-4.amzn2023.0.3                  amazonlinux   717 k
dejavu-sans-fonts                      noarch    2.37-16.amzn2023.0.2                   amazonlinux   1.3 M
dejavu-sans-mono-fonts                 noarch    2.37-16.amzn2023.0.2                   amazonlinux   467 k
dejavu-serif-fonts                     noarch    2.37-16.amzn2023.0.2                   amazonlinux   1.0 M
fontconfig                             x86_64    2.13.94-2.amzn2023.0.2                 amazonlinux   273 k
fonts-filesystem                       noarch    1:2.0.5-12.amzn2023.0.2                 amazonlinux   9.5 k
freetype                               x86_64    2.13.2-5.amzn2023.0.2                  amazonlinux   422 k
giflib                                 x86_64    5.2.1-9.amzn2023.0.3                   amazonlinux   48 k
google-noto-fonts-common                noarch    20240401-1.amzn2023.0.2                 amazonlinux   17 k
google-noto-sans-vf-fonts               noarch    20240401-1.amzn2023.0.2                 amazonlinux   593 k
graphite2                               x86_64    1.3.14-7.amzn2023.0.2                  amazonlinux   97 k
harfbuzz                               x86_64    7.0.0-2.amzn2023.0.2                   amazonlinux   873 k
java-21-amazon-corretto-headless        x86_64    1:21.0.11+10-1.amzn2023.1            amazonlinux   97 M
javapackages-filesystem                 noarch    6.0.0-7.amzn2023.0.6                   amazonlinux   12 k
langpacks-core-font-en                  noarch    3.0-21.amzn2023.0.4                    amazonlinux   10 k
libICE                                  x86_64    1.1.1-3.amzn2023.0.1                   amazonlinux   76 k
libSM                                   x86_64    1.2.4-3.amzn2023.0.1                   amazonlinux   45 k
libX11                                  x86_64    1.8.10-2.amzn2023.0.1                  amazonlinux   659 k
```

```
root@ip-172-31-39-19:/home/ec2-user

google-noto-sans-vf-fonts-20240401-1.amzn2023.0.2.noarch
graphite2-1.3.14-7.amzn2023.0.2.x86_64
harfbuzz-7.0.0-2.amzn2023.0.2.x86_64
java-21-amazon-corretto-1:21.0.11+10-1.amzn2023.1.x86_64
java-21-amazon-corretto-headless-1:21.0.11+10-1.amzn2023.1.x86_64
javapackages-filesystem-6.0.0-7.amzn2023.0.6.noarch
langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch
libICE-1.1.1-3.amzn2023.0.1.x86_64
libSM-1.2.4-3.amzn2023.0.1.x86_64
libX11-1.8.10-2.amzn2023.0.1.x86_64
libX11-common-1.8.10-2.amzn2023.0.1.noarch
libXau-1.0.11-6.amzn2023.0.1.x86_64
libXext-1.3.6-1.amzn2023.0.1.x86_64
libXi-1.8.2-1.amzn2023.0.1.x86_64
libXinerama-1.1.5-6.amzn2023.0.1.x86_64
libXrandr-1.5.4-3.amzn2023.0.1.x86_64
libXrender-0.9.11-6.amzn2023.0.1.x86_64
libXt-1.3.0-3.amzn2023.0.1.x86_64
libXtst-1.2.5-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
libpng-2:1.6.37-10.amzn2023.0.12.x86_64
libxcb-1.17.0-1.amzn2023.0.1.x86_64
pixman-0.43.4-1.amzn2023.0.4.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
[root@ip-172-31-39-19 ec2-user]#
```

Install Jenkins

```
root@ip-172-31-39-19:/home/ec2-user
[root@ip-172-31-39-19 ec2-user]# sudo yum install java-21-amazon-corretto -y
Last metadata expiration check: 0:01:05 ago on Mon May 11 09:48:30 2026.
Dependencies resolved.

=====
Package                                Arch      Version                                Repository      Size
=====
Installing:
java-21-amazon-corretto                x86_64    1:21.0.11+10-1.amzn2023.1            amazonlinux     218 k
Installing dependencies:
alsa-lib                               x86_64    1.2.7.2-1.amzn2023.0.3                amazonlinux     503 k
cairo                                   x86_64    1.18.0-4.amzn2023.0.3                 amazonlinux     717 k
dejavu-sans-fonts                       noarch    2.37-16.amzn2023.0.2                  amazonlinux     1.3 M
dejavu-sans-mono-fonts                  noarch    2.37-16.amzn2023.0.2                  amazonlinux     467 k
dejavu-serif-fonts                      noarch    2.37-16.amzn2023.0.2                  amazonlinux     1.0 M
fontconfig                              x86_64    2.13.94-2.amzn2023.0.2                amazonlinux     273 k
fonts-filesystem                       noarch    1:2.0.5-12.amzn2023.0.2               amazonlinux     9.5 k
freetype                                x86_64    2.13.2-5.amzn2023.0.2                 amazonlinux     422 k
giflib                                  x86_64    5.2.1-9.amzn2023.0.3                  amazonlinux     48 k
google-noto-fonts-common                noarch    20240401-1.amzn2023.0.2               amazonlinux     17 k
google-noto-sans-vf-fonts               noarch    20240401-1.amzn2023.0.2               amazonlinux     593 k
graphite2                               x86_64    1.3.14-7.amzn2023.0.2                 amazonlinux     97 k
harfbuzz                                x86_64    7.0.0-2.amzn2023.0.2                  amazonlinux     873 k
java-21-amazon-corretto-headless        x86_64    1:21.0.11+10-1.amzn2023.1            amazonlinux     97 M
javapackages-filesystem                 noarch    6.0.0-7.amzn2023.0.6                  amazonlinux     12 k
langpacks-core-font-en                  noarch    3.0-21.amzn2023.0.4                   amazonlinux     10 k
libICE                                   x86_64    1.1.1-3.amzn2023.0.1                  amazonlinux     76 k
libSM                                    x86_64    1.2.4-3.amzn2023.0.1                  amazonlinux     45 k
libX11                                   x86_64    1.8.10-2.amzn2023.0.1                 amazonlinux     659 k
=====
```

```
root@ip-172-31-39-19:/home/ec2-user
jenkins                                noarch    2.555.1-1                             jenkins         95 M

Transaction Summary
=====
Install 1 Package

Total download size: 95 M
Installed size: 96 M
Downloading Packages:
jenkins-2.555.1-1.noarch.rpm            10 MB/s | 95 MB | 00:09
-----
Total                                  10 MB/s | 95 MB | 00:09
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : jenkins-2.555.1-1.noarch                    1/1
  Running scriptlet: jenkins-2.555.1-1.noarch                    1/1
  Installing     : jenkins-2.555.1-1.noarch                    1/1
  Running scriptlet: jenkins-2.555.1-1.noarch                    1/1
  Verifying      : jenkins-2.555.1-1.noarch                    1/1

Installed:
jenkins-2.555.1-1.noarch

Complete!
[root@ip-172-31-39-19 ec2-user]#
```

Changed the port number from 8080 to 8081


```
root@ip-172-31-39-19:/home/ec2-user
GNU nano 8.3 /lib/systemd/system/jenkins.service Modified
#Environment="JENKINS_LOG=%L/jenkins/jenkins.log"

# The Java home directory. When left empty, JENKINS_JAVA_CMD and PATH are consulted.
#Environment="JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64"

# The Java executable. When left empty, JAVA_HOME and PATH are consulted.
#Environment="JENKINS_JAVA_CMD=/etc/alternatives/java"

# Arguments for the Jenkins JVM
Environment="JAVA_OPTS=-Djava.awt.headless=true"

# Unix Domain Socket to listen on for local HTTP requests. Default is disabled.
#Environment="JENKINS_UNIX_DOMAIN_PATH=/run/jenkins/jenkins.socket"

# IP address to listen on for HTTP requests.
# The default is to listen on all interfaces (0.0.0.0).
#Environment="JENKINS_LISTEN_ADDRESS="

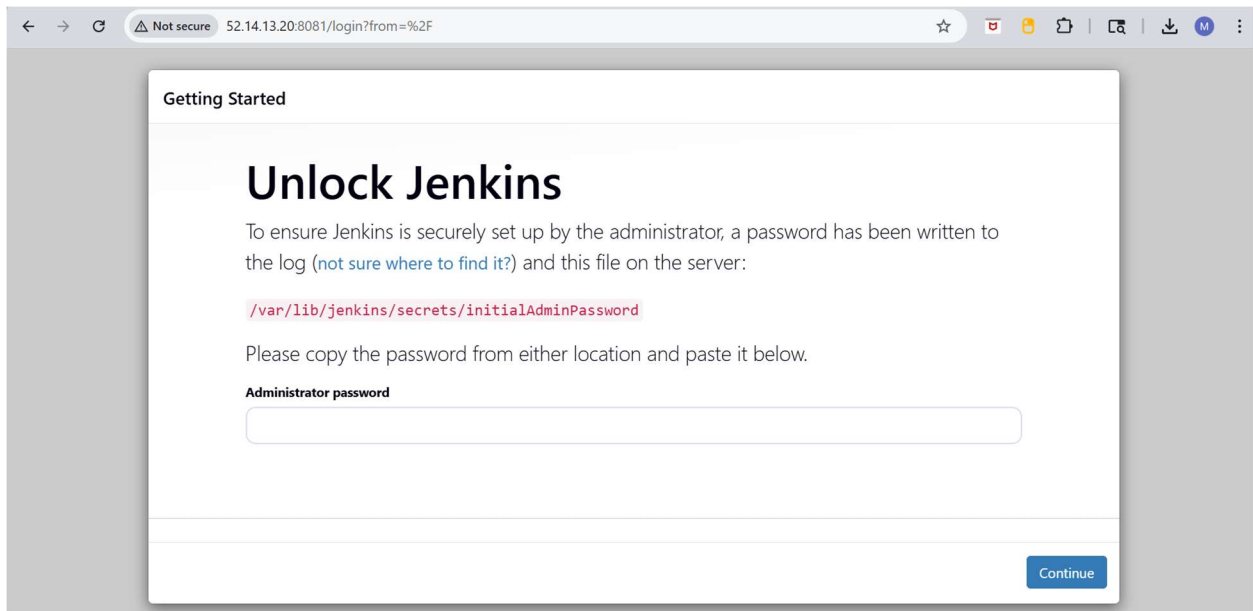
# Port to listen on for HTTP requests. Set to -1 to disable.
# To be able to listen on privileged ports (port numbers less than 1024),
# add the CAP_NET_BIND_SERVICE capability to the AmbientCapabilities
# directive below.
Environment="JENKINS_PORT=8081"

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_/ Go To Line

[root@ip-172-31-39-19 ec2-user]# sudo systemctl daemon-reload
[root@ip-172-31-39-19 ec2-user]# sudo systemctl start jenkins
[root@ip-172-31-39-19 ec2-user]# sudo systemctl enable jenkins
[root@ip-172-31-39-19 ec2-user]# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-05-11 09:57:16 UTC; 2min 26s ago
     Main PID: 26521 (java)
        Tasks: 37 (limit: 1014)
      Memory: 274.3M
         CPU: 15.999s
    CGroup: /system.slice/jenkins.service
            └─26521 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.wa

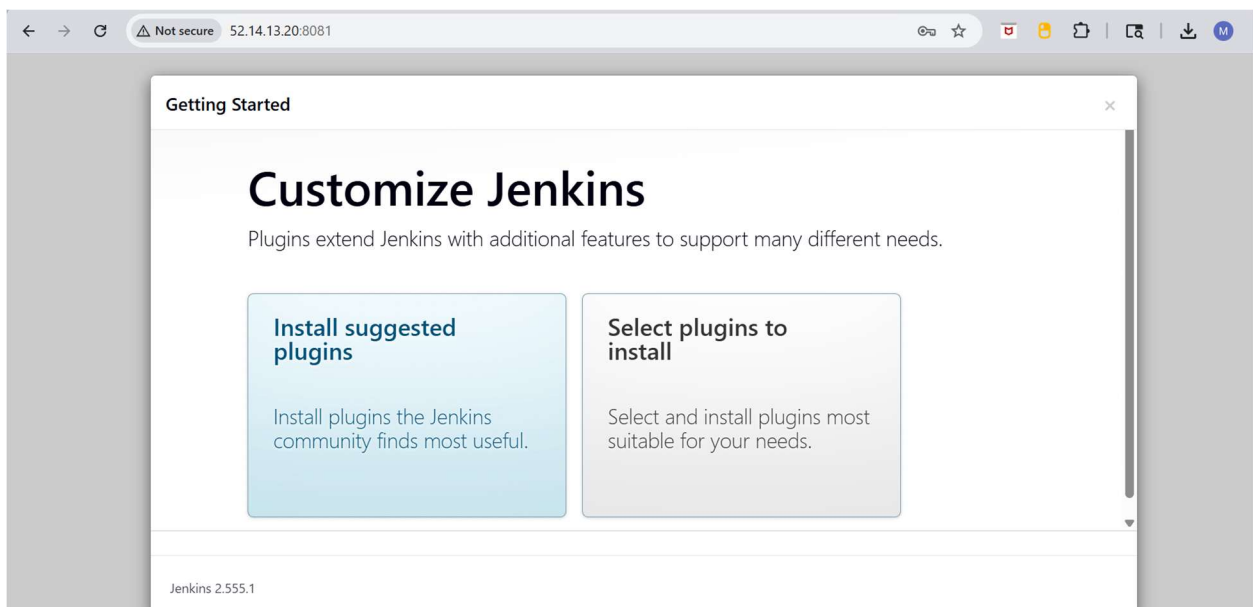
May 11 09:57:12 ip-172-31-39-19.us-east-2.compute.internal jenkins[26521]: [LF]> *****
May 11 09:57:12 ip-172-31-39-19.us-east-2.compute.internal jenkins[26521]: [LF]> *****
May 11 09:57:16 ip-172-31-39-19.us-east-2.compute.internal jenkins[26521]: 2026-05-11 09:57
May 11 09:57:16 ip-172-31-39-19.us-east-2.compute.internal jenkins[26521]: 2026-05-11 09:57
May 11 09:57:21 ip-172-31-39-19.us-east-2.compute.internal systemd[1]: Started jenkins.serv
May 11 09:58:03 ip-172-31-39-19.us-east-2.compute.internal jenkins[26521]: 2026-05-11 09:57
May 11 09:58:03 ip-172-31-39-19.us-east-2.compute.internal systemd[1]: /usr/lib/systemd/sys
May 11 09:59:26 ip-172-31-39-19.us-east-2.compute.internal systemd[1]: /usr/lib/systemd/sys
May 11 09:59:26 ip-172-31-39-19.us-east-2.compute.internal systemd[1]: /usr/lib/systemd/sys
May 11 09:59:42 ip-172-31-39-19.us-east-2.compute.internal systemd[1]: /usr/lib/systemd/sys
lines 1-20/20 (END)
```

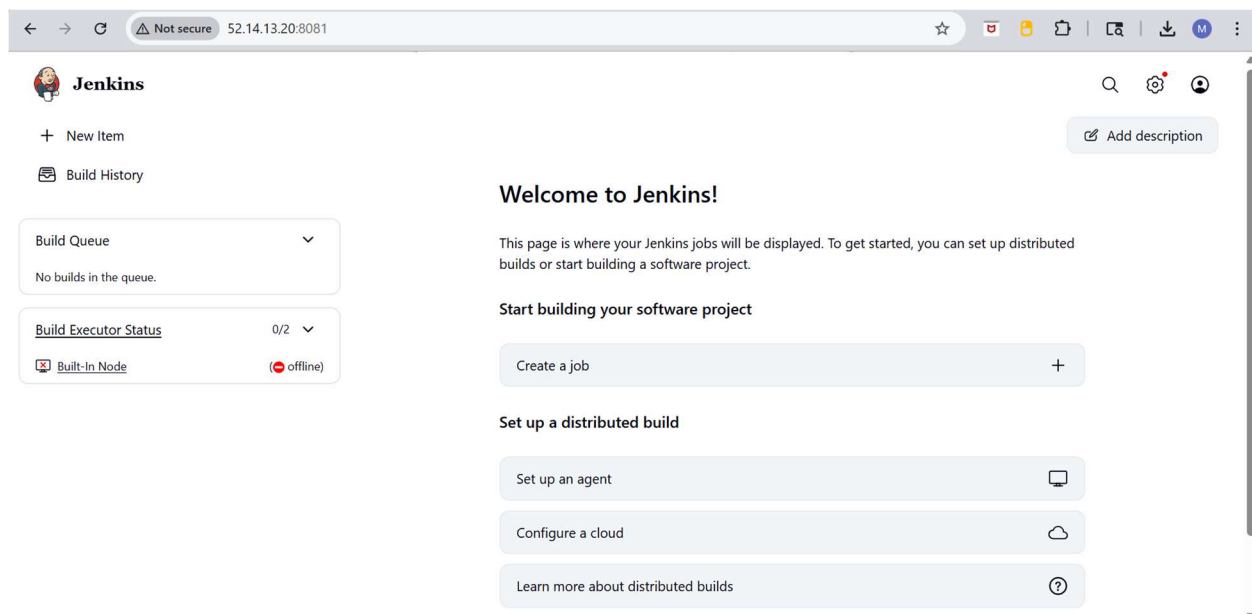
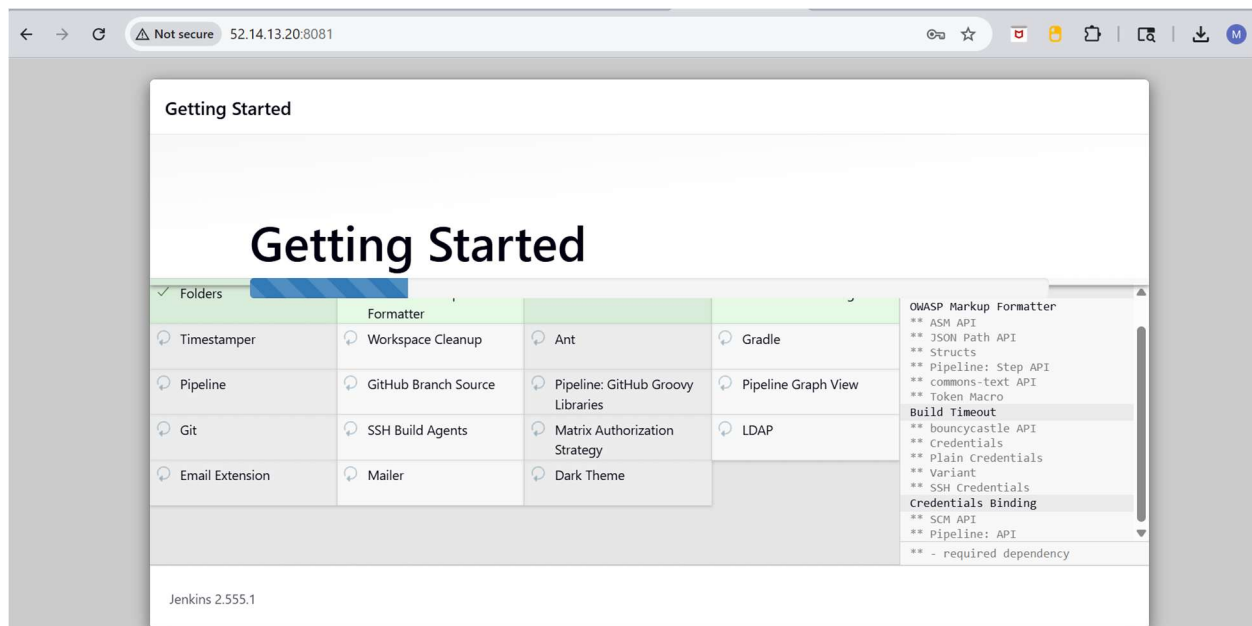
Verified the browser



Login with the admin password

Click on install suggested plugins





2. Secure Jenkins server

I have changed the password to secure Jenkins server

← → ↻ ⚠ Not secure 52.14.13.20:8081 🔒 ☆ 📄 🗂 📎 📌 📁 M ⋮

Getting Started

Create First Admin User

Username

Password

Confirm password

Jenkins 2.555.1

[Skip and continue as admin](#) [Save and Continue](#)

← → ↻ ⚠ Not secure 52.14.13.20:8081 🔒 ☆ 📄 🗂 📎 📌 📁 M ⋮

Getting Started

Instance Configuration

Jenkins URL:

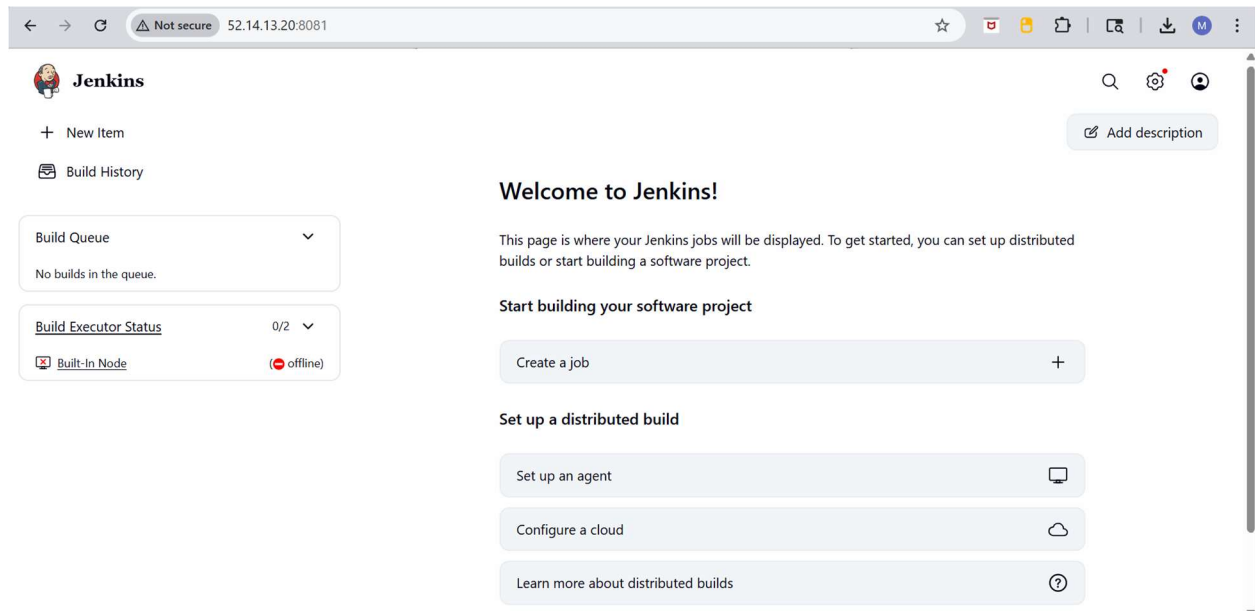
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Jenkins 2.555.1

[Not now](#) [Save and Finish](#)

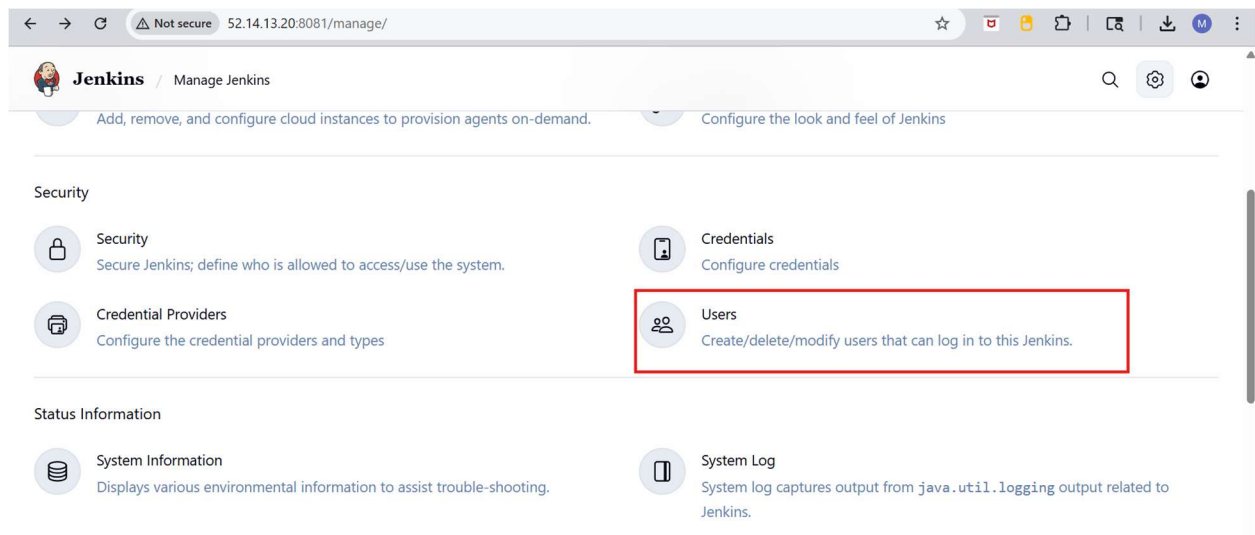
Successfully changed password and logged into jenkins



3.Create users called DevOps, Testing in Jenkins with Limited access.

Create user

Navigate to the Manage Jenkins page from the Jenkins dashboard → Select Users under the Security section.



To create a users

On the Users page, select Create User.

← → ↻ ⚠ Not secure 52.14.13.20:8081/manage/securityRealm/ ☆ 📄 🗂 📏 📄 ⬇ M ⋮

 **Jenkins** / Manage Jenkins ▾ / Jenkins' own user database 🔍 ⚙️ 👤

Users 1 + Create User

These users can log into Jenkins. This is a sub set of [this list](#), which also contains auto-created users who really just made some commits on some projects and have no direct Jenkins access.

User ID 1	Name
 admin	admin 

Fill in the required details:

- **Username:** A unique identifier for the user.
- **Password:** A secure password for the user.
- **Confirm Password:** Re-enter the password.
- **Full Name:** The full name of the user.
- **Email Address:** The email address of the user.

← → ↻ ⚠ Not secure 52.14.13.20:8081/manage/securityRealm/addUser 🔍 ⚙️ 👤

 **Jenkins** / Manage Jenkins ▾ / Jenkins' own user database / Create User 🔍 ⚙️ 👤

Create User

Username

Password

Confirm password

Full name

E-mail address

Create User

← → ↻ ⚠ Not secure 52.14.13.20:8081/manage/securityRealm/addUser 🔒 ☆ 📄 🗂 🔍 👤

Jenkins / Manage Jenkins ▾ / Jenkins' own user database / Create User 🔍 ⚙️ 👤

Create User

Username

Password

Confirm password

Full name

E-mail address

[Create User](#)

Go to manage Jenkins → click on plugins

← → ↻ ⚠ Not secure 52.14.13.20:8081/manage/ ☆ 📄 🗂 🔍 👤

Jenkins / Manage Jenkins 🔍 ⚙️ 👤

Manage Jenkins

🔍 Search settings /

System Configuration

**System**
Configure global settings and paths.

**Tools**
Configure tools, their locations and automatic installers.

**Plugins**
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.

**Nodes**
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

**Clouds**
Add, remove, and configure cloud instances to provision agents on-demand.

**Appearance**
Configure the look and feel of Jenkins

Security

Click on available plugins

← → ↻ ⚠ Not secure 52.14.13.20:8081/manage/pluginManager/

 **Jenkins** / Manage Jenkins ▾ / Plugins

Plugins

🔍 Search plugin updates

- 📁 Updates
- 📁 Available plugins**
- ⚙ Installed plugins
- ⚙ Advanced settings
- ☰ Download progress

Search Role-based authorization strategy and install

← → ↻ ⚠ Not secure 52.14.13.20:8081/manage/pluginManager/available

 **Jenkins** / Manage Jenkins ▾ / Plugins 🔍 ⚙ 👤

Plugins

🔍 Role-based Authorization Strategy

📁 Updates

📁 Available plugins

⚙ Installed plugins

⚙ Advanced settings

☰ Download progress

Install	Name	Released	Health
☒	Role-based Authorization Strategy 848.va_a_ea_673cf0b_c Security Authentication and User Management Enables user authorization using a Role-Based strategy. Roles can be defined globally or for particular jobs or nodes selected by regular expressions.	5 mo 9 days ago	97

← → ↻ Not secure 52.14.13.20:8081/manage/pluginManager/updates/ ☆ 📄 📁 📄 📄 📄 M ⋮

Jenkins / Manage Jenkins ▾ / Plugins 🔍 ⚙️ 👤

Plugins

- 📁 Updates
- 📁 Available plugins
- 🔧 Installed plugins
- ⚙️ Advanced settings
- ☰ Download progress**

Download progress

Preparation	
	• Checking internet connectivity
	• Checking update center connectivity
	• Success
commons-lang3 v3.x Jenkins API	✓ Success
Ionicons API	✓ Success
Folders	✓ Success
OWASP Markup Formatter	✓ Success
ASM API	✓ Success
JSON Path API	✓ Success
Structs	✓ Success
Pipeline: Step API	✓ Success
commons-text API	✓ Success
Token Macro	✓ Success
Build Timeout	✓ Success
hudsonette API	✓ Success

Enable project based on Jenkins

Jenkins / Manage Jenkins 🔍 ⚙️ 👤

 **Clouds**
Add, remove, and configure cloud instances to provision agents on-demand.

 **Appearance**
Configure the look and feel of Jenkins

Security

 **Security**
Secure Jenkins; define who is allowed to access/use the system.

 **Credential Providers**
Configure the credential providers and types

 **Credentials**
Configure credentials

 **Users**
Create/delete/modify users that can log in to this Jenkins.

Status Information

 **System Information**
Displays various environmental information to assist trouble-shooting.

 **System Log**
System log captures output from java.util.logging output related to Jenkins.

 **Load Statistics**

 **About Jenkins**

Under authorization select project based matrix authorization strategy

← → ↻ Not secure 52.14.13.20:8081/manage/configureSecurity/

Jenkins / Manage Jenkins / Security

Authorization

Project-based Matrix Authorization Strategy

User/group	Overall	Credentials				Agent				Job				Run		View																
	Administer	Manage	Read	Create	Delete	ManageDomains	Update	View	Build	Configure	Connect	Create	Delete	Disconnect	Provision	Build	Cancel	Configure	Create	Delete	Discover	Move	Read	Workspace	Delete	Replay	Update	Configure	Create	Delete	Read	
Anonymous																																
Authenticated Users																																
DevOps-19																																
Testing-19																																

Add user... Add group... ?

Save Apply

Assign roles

← → ↻ Not secure 52.14.13.20:8081/manage/configureSecurity/

Jenkins / Manage Jenkins / Security

Project-based Matrix Authorization Strategy

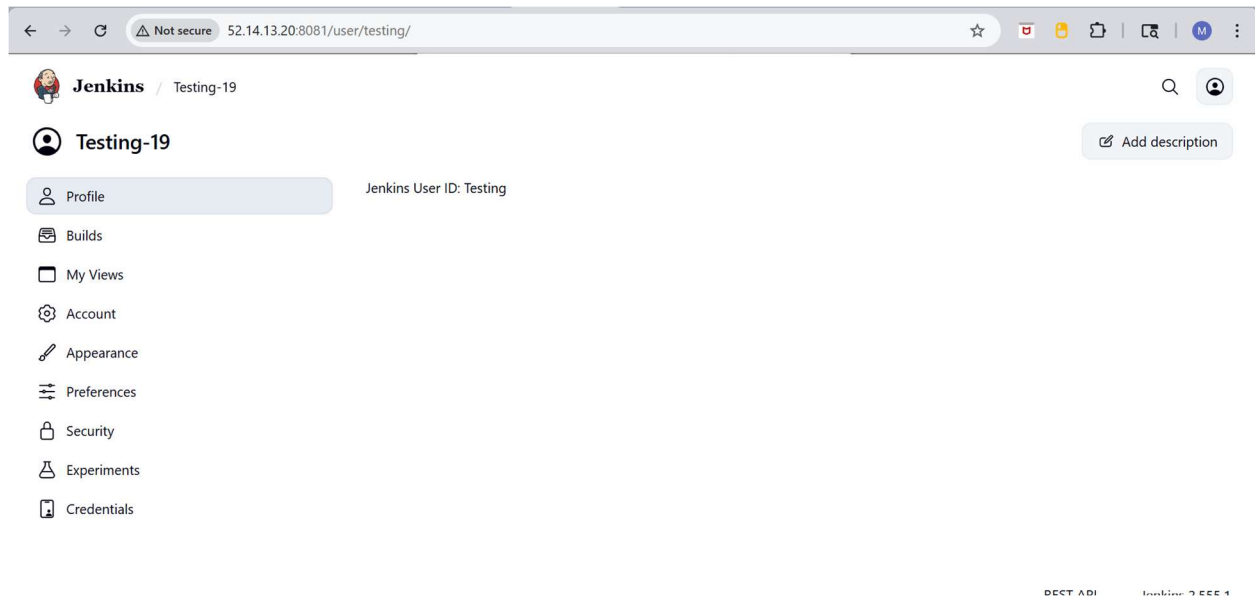
User/group	Overall	Credentials				Agent				Job				Run		View		SCM	Meta															
	Administer	Manage	Read	Create	Delete	ManageDomains	Update	View	Build	Configure	Connect	Create	Delete	Disconnect	Provision	Build	Cancel	Configure	Create	Delete	Discover	Move	Read	Workspace	Delete	Replay	Update	Configure	Create	Delete	Read	Tag	HealthCheck	InnreadDump
Anonymous																																		
Authenticated Users																																		
admin	✓																																	
DevOps-19			✓	✓							✓								✓					✓				✓			✓			
Testing-19			✓																				✓					✓			✓			

Add user... Add group... ?

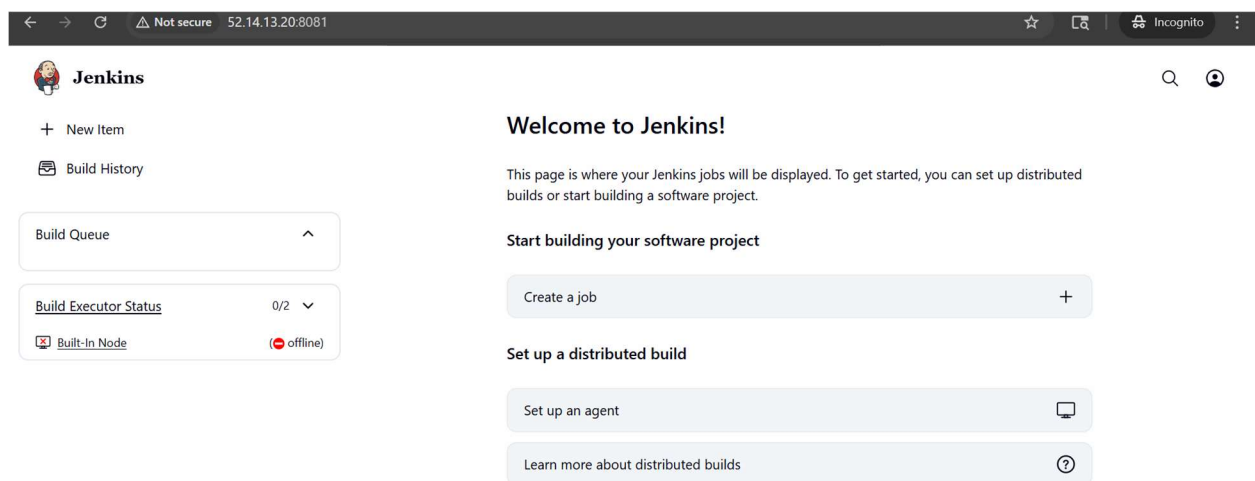
Save Apply

Tried verifying the user

Below for user – Testing I am unable to create as I only have read access



In DevOps user I am able to create a job as I gave create access



4. Configure labels and restrict the jobs to execute based on label only.

Go to manage Jenkins → click on nodes

← → ↻ ⚠ Not secure 18.221.133.148:8081/manage/ ☆ 🇮🇹 📁 🗑️ 🔍 👤

Jenkins / Manage Jenkins 🔍 ⚙️ 👤

Manage Jenkins

🔍 Search settings /

System Configuration

**System**
Configure global settings and paths.

**Tools**
Configure tools, their locations and automatic installers.

**Plugins**
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.

**Nodes**
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

**Clouds**
Add, remove, and configure cloud instances to provision agents on-demand.

**Appearance**
Configure the look and feel of Jenkins

Security

**Security**
Secure Jenkins; define who is allowed to access/use the system.

**Credentials**
Configure credentials

Click on built in node

← → ↻ ⚠ Not secure 18.221.133.148:8081/computer/(built-in)/configure ☆ 🇮🇹 📁 🗑️ 🔍 👤

Jenkins / Nodes ▾ / Built-In Node ▾ / Configure 🔍 ⚙️ 👤

**Status**

**Configure**

**Build History**

**Load Statistics**

**Script Console**

Build Executor Status 0/2 ▾

Number of executors ?
2

Labels ?
dev

Usage ?
Use this node as much as possible ▾

Node Properties

☐ Disable deferred wipeout on this node ?

☐ Disk Space Monitoring Thresholds

☐ Enable node-based security

Save **Apply**

Click on new item → give item name → choose item type



New Item

Enter an item name

dev

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK



Configure

⚙️ General

🔗 Source Code Management

🕒 Triggers

🌐 Environment

📋 Build Steps

📦 Post-build Actions

☐ This project is parameterized ?

☐ Throttle builds ?

☐ Execute concurrent builds if necessary ?

☒ Restrict where this project can be run ?

Label Expression ?

dev

Label dev matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced ▾

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

Save

Apply

Jenkins / dev / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See the list of available environment variables

```
sleep 60
```

Advanced ▾

Save Apply

Jenkins

+ New Item

Build History

Build Queue ▾
No builds in the queue.

Build Executor Status 0/2 ▾

All +

S	W	Name 1	Last Success	Last Failure	Last Duration
✓	☀	dev	14 hr #1	N/A	1 min 0 sec

5. Create Three sample jobs using the below URL.

Source code: https://github.com/betawins/Techie_horizon_Login_app

- Install git plugins in Jenkins server
- Go to Manage Jenkins → Plugins → available plugins

Jenkins / Manage Jenkins / Plugins

Plugins

Search: bitbucket

Install ▾

Install	Name 1	Released	Health
☒	Bitbucket 307.va_9f30d0b_8f0d Integrates with BitBucket	1 mo 1 day ago	100
☐	Bitbucket Branch Source 937.3.1 bitbucket Source Code Management Allows to use Bitbucket Cloud and Bitbucket Server as sources for multi-branch projects. It also provides the required connectors for Bitbucket Cloud	25 days ago	94

← → ↻ Not secure 3.133.58.30:8081/manage/pluginManager/updates/ ☆ 📄 🗑️ 🔍 M ⋮

 **Jenkins** / Manage Jenkins ▾ / Plugins 🔍 ⚙️ 👤

Plugins

- 📄 Updates
- 📁 Available plugins
- 🔧 Installed plugins
- ⚙️ Advanced settings
- ☰ Download progress**

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

JSch dependency ✓ Success

Mercurial ✓ Success

Bitbucket ✓ Success

Loading plugin extensions ✓ Success

→ [Go back to the top page](#)
(you can start using the installed plugins right away)

→ ☐ Restart Jenkins when installation is complete and no jobs are running

```

root@ip-172-31-39-19:/home/ec2-user
[root@ip-172-31-39-19 ec2-user]# sudo yum install git
Amazon Linux 2023 Kernel Livepatch repository 333 kB/s | 38 kB 00:00
Jenkins-stable 10 kB/s | 1.9 kB 00:00
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
git x86_64 2.50.1-1.amzn2023.0.1 amazonlinux 53 k
Installing dependencies:
git-core x86_64 2.50.1-1.amzn2023.0.1 amazonlinux 4.9 M
git-core-doc noarch 2.50.1-1.amzn2023.0.1 amazonlinux 2.8 M
perl-Error noarch 1:0.17030-2.amzn2023.0.1 amazonlinux 42 k
perl-File-Find noarch 1.37-477.amzn2023.0.7 amazonlinux 25 k
perl-Git noarch 2.50.1-1.amzn2023.0.1 amazonlinux 41 k
perl-TermReadKey x86_64 2.38-9.amzn2023.0.2 amazonlinux 36 k
perl-lib x86_64 0.65-477.amzn2023.0.7 amazonlinux 15 k
Transaction Summary
=====
Install 8 Packages

Total download size: 7.9 M
Installed size: 41 M

```

Create new item

Jenkins / job-1 / Configure

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/betawins/Techie_horizon_Login_app

Credentials ?

- none - + Add

Advanced ▾

Save Apply

Jenkins / job-1 / Build Time Trend

Build Time Trend

- Status
- Changes
- Workspace
- Build Now
- Configure
- Delete Project
- Rename

Builds > Filter /

Today

✓ #1 4:30 am



S Build 1 Time Since Duration

Open console to check the status

 Jenkins

job-1 / #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

Console

DownloadCopyView as plain text

Started by user admin

Running as SYSTEM

Building in workspace /var/lib/jenkins/workspace/job-1

The recommended git tool is: NONE

No credentials specified

Cloning the remote Git repository

Cloning repository https://github.com/betawins/Techie_horizon_Login_app

> git init /var/lib/jenkins/workspace/job-1 # timeout=10

Fetching upstream changes from https://github.com/betawins/Techie_horizon_Login_app

> git --version # timeout=10

> git --version # 'git version 2.50.1'

> git fetch --tags --force --progress -- https://github.com/betawins/Techie_horizon_Login_app +refs/heads/*:refs/remotes/origin/* # timeout=10

> git config remote.origin.url https://github.com/betawins/Techie_horizon_Login_app # timeout=10

> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10

Avoid second fetch

> git rev-parse refs/remotes/origin/master^{commit} # timeout=10

Checking out Revision ef9c453d4993843ac49d37321316c2588f154495 (refs/remotes/origin/master)

> git config core.sparsecheckout # timeout=10

Lets create 2 more jobs

 Jenkins

All / New Item

SearchSettingsUser

Enter an item name

job-2

Select an item type

Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view,

OK

← → ↻ ⚠ Not secure 3.133.58.30:8081/job/job-2/configure ☆ 📄 🗑 📄 📄 📄 📄

Jenkins / job-2 / Configure 🔍 ⚙️ 👤

Configure

🔧 General

🔑 Source Code Management

🕒 Triggers

🌐 Environment

📋 Build Steps

📦 Post-build Actions

None

Git ?

Repositories ?

Repository URL ?

https://github.com/betawins/Techie_horizon_Login_app.git

Credentials ?

- none - + Add

Advanced ▾

+ Add Repository

Save Apply

Jenkins / job-2 / #1 🔍 ⚙️ 👤

Status

</> Changes

📄 Console Output

📄 Edit Build Information

🗑 Delete build '#1'

🕒 Timings

🔑 Git Build Data

✅ **#1 (12-May-2026, 6:14:10 am)** 📄 Add description Keep this build forever

🕒 Started by user [admin](#) Started 9.7 sec ago Took 0.38 sec

🕒 This run spent:

- 4 ms waiting;
- 0.38 sec build duration;
- 0.38 sec total from scheduled to completion.

🔑 **git** **Revision:** [ef9c453d4993843ac49d37321316c2588f154495](#) **Repository:** https://github.com/betawins/Techie_horizon_Login_app.git

- refs/remotes/origin/master

</> No changes.

REST API Jenkins 2.555.1

Open console to check the status

Jenkins / job-2 / #2

Delete build '#2'

Timings

Git Build Data

Previous Build

```
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/job-2/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/betawins/Techie_horizon_Login_app.git # timeout=10
Fetching upstream changes from https://github.com/betawins/Techie_horizon_Login_app.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/betawins/Techie_horizon_Login_app.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision ef9c453d4993843ac49d37321316c2588f154495 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f ef9c453d4993843ac49d37321316c2588f154495 # timeout=10
Commit message: "sabit code for testing"
> git rev-list --no-walk ef9c453d4993843ac49d37321316c2588f154495 # timeout=10
Finished: SUCCESS
```

Jenkins / All / New Item

New Item

Enter an item name

job-3

Select an item type

Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

Jenkins / job-3 / #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

#1 (12-May-2026, 6:21:22 am)

Started by user admin

This run spent:

- 5 ms waiting;
- 0.35 sec build duration;
- 0.36 sec total from scheduled to completion.

git

Revision: ef9c453d4993843ac49d37321316c2588f154495

Repository: https://github.com/betawins/Techie_horizon_Login_app.git

- refs/remotes/origin/master

</>

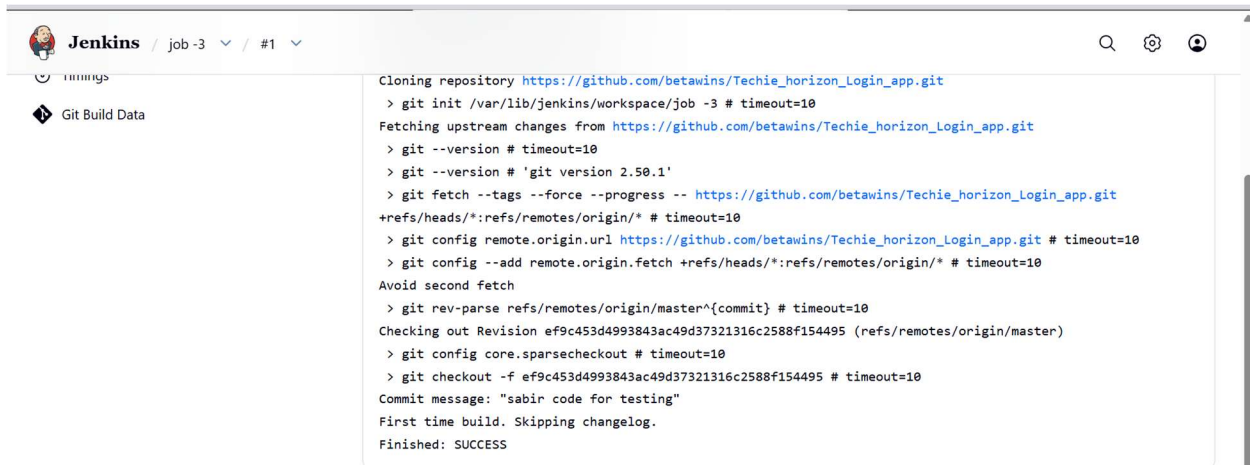
No changes.

Add description

Keep this build forever

Started 13 sec ago

Took 0.35 sec

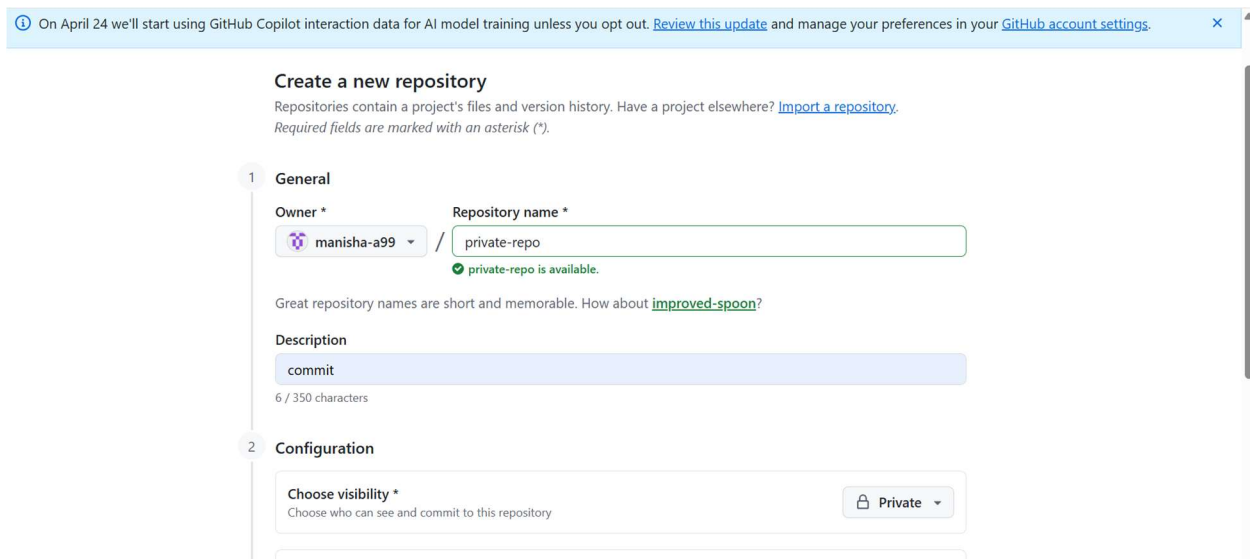


The screenshot shows the Jenkins web interface for a job named 'job-3'. The console output displays a series of git commands used to clone a repository from GitHub, fetch updates, and checkout a specific commit. The build concludes with a 'SUCCESS' status.

```
Cloning repository https://github.com/betawins/Techie_horizon_Login_app.git
> git init /var/lib/jenkins/workspace/job -3 # timeout=10
Fetching upstream changes from https://github.com/betawins/Techie_horizon_Login_app.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/betawins/Techie_horizon_Login_app.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/betawins/Techie_horizon_Login_app.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision ef9c453d4993843ac49d37321316c2588f154495 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f ef9c453d4993843ac49d37321316c2588f154495 # timeout=10
Commit message: "sabir code for testing"
First time build. Skipping changelog.
Finished: SUCCESS
```

6.Create one Jenkins job using GitHub Private repository.

Go to github and create new repository



The screenshot shows the GitHub 'Create a new repository' page. The 'General' tab is active, showing the repository owner as 'manisha-a99' and the repository name as 'private-repo'. A green checkmark indicates that the repository is available. The description field contains the text 'commit'. The 'Configuration' tab is also visible, showing the visibility set to 'Private'.

On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. [Review this update](#) and manage your preferences in your [GitHub account settings](#).

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

1 General

Owner * / Repository name *

private-repo is available.

Great repository names are short and memorable. How about [improved-spoon](#)?

Description

6 / 350 characters

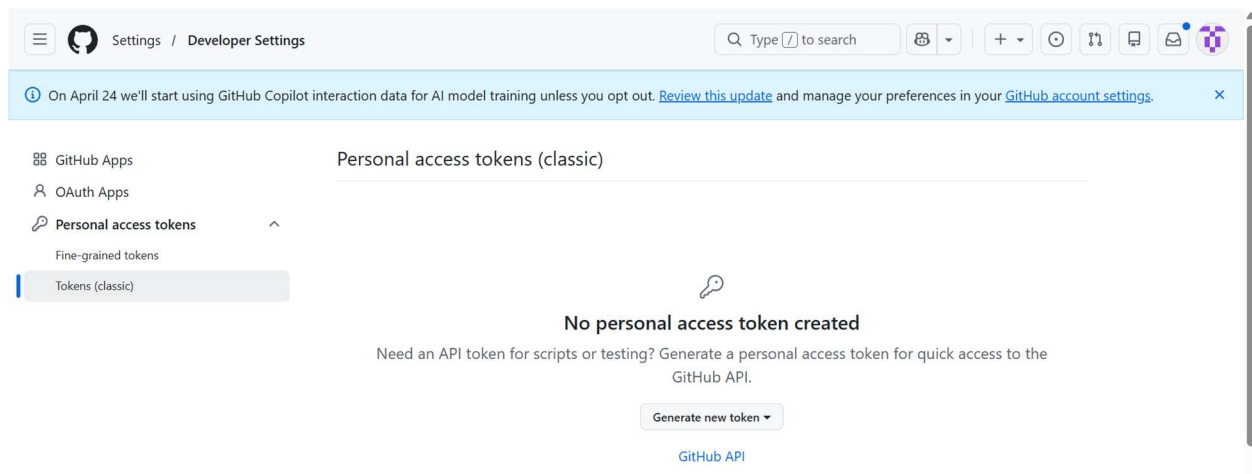
2 Configuration

Choose visibility *

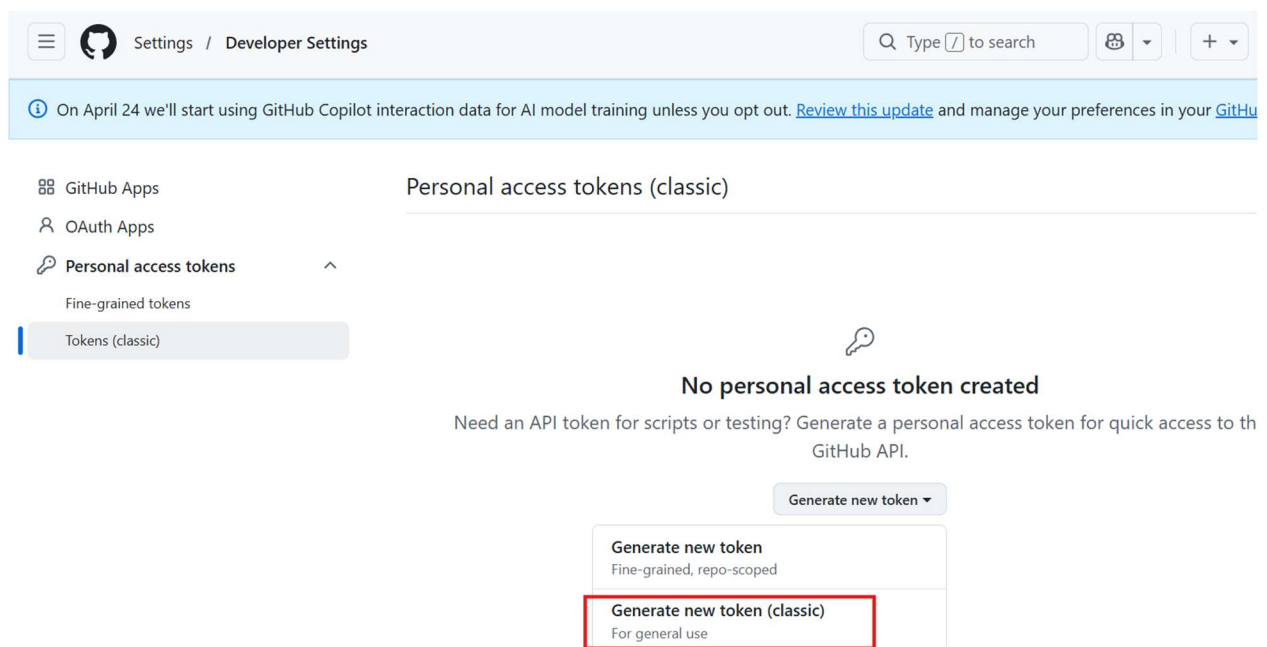
Choose who can see and commit to this repository

Generate token

Go to settings → go to developers → token



Click on generate new token classic



Given name in note and enable repo and generate token and kept safely.

GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

jenkins-01

What's this token for?

Expiration

30 days (Jun 11, 2026)

The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☒ repo

Full control of private repositories

Go to manage Jenkins → credentials

JenkinsManage Jenkins

Add, remove, disable or enable plugins that can extend the functionality of Jenkins.

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

Clouds

Add, remove, and configure cloud instances to provision agents on-demand.

Appearance

Configure the look and feel of Jenkins

Security

Security

Secure Jenkins; define who is allowed to access/use the system.

Credentials

Configure credentials

Users

Create/delete/modify users that can log in to this Jenkins.

Credential Providers

Configure the credential providers and types

Select Global and Click Add Credentials and select username and password

JenkinsManage Jenkins

Global

Credentials that should be available everywhere

Add Credentials

Select a type of credential

Username with password

Commonly used for authentication to services like Git, APIs, or registries.

GitHub App

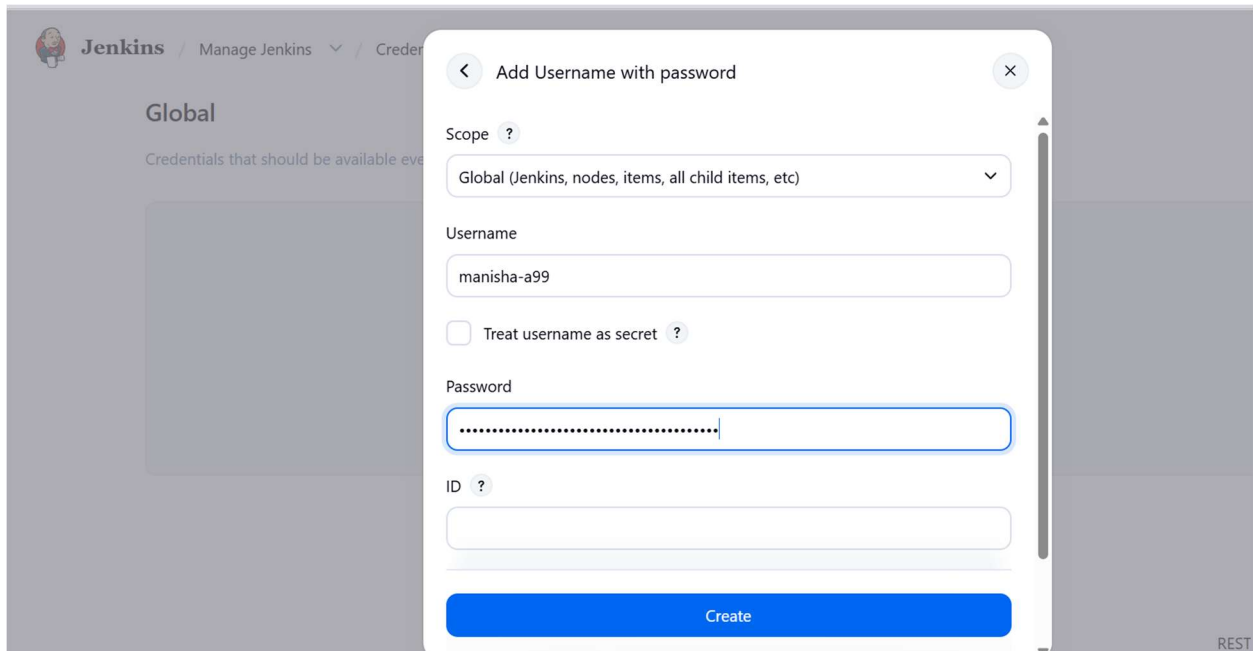
SSH Username with private key

Secret file

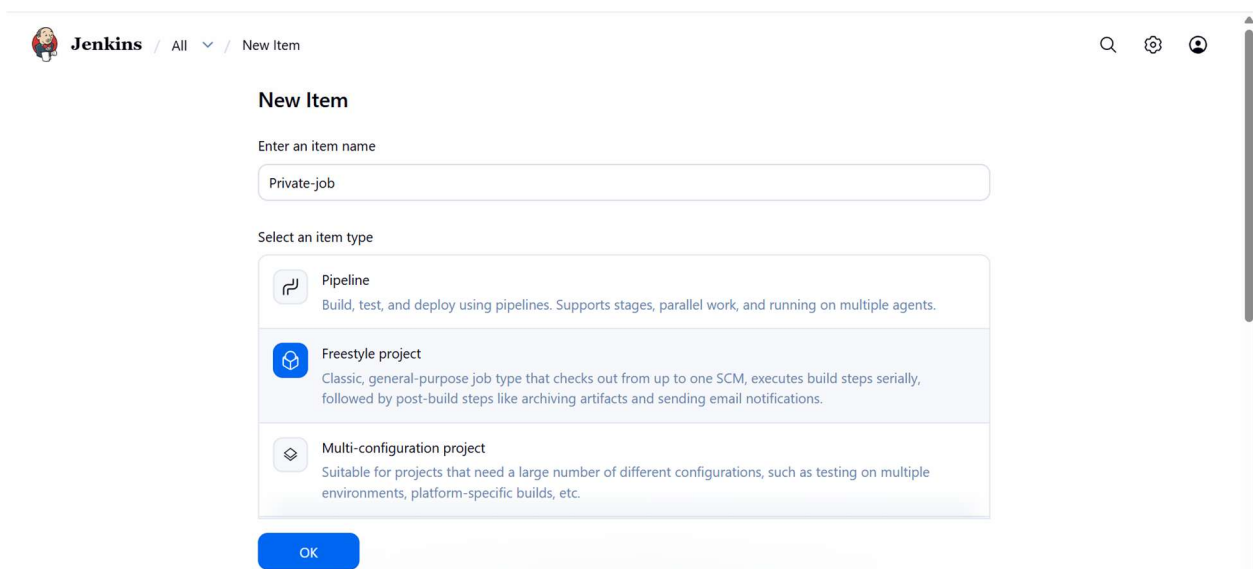
Secret text

Next

REST APIJenkins 2.555.1



Create new item



In Source Management change git and added my private git URL

 Jenkins

Private-job

Configure

Q

⚙️

👤

Configure

⚙️ General

🔑 Source Code Management

🕒 Triggers

🌐 Environment

📋 Build Steps

🏠 Post-build Actions

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git

Repositories

Repository URL

https://github.com/manisha-a99/private-repo.git

Credentials

manisha-a99/*****

+ Add

Advanced

Save

Apply

Create a build

 Jenkins

private-build

Configure

Q

⚙️

👤

Configure

⚙️ General

🔑 Source Code Management

🕒 Triggers

🌐 Environment

📋 Build Steps

🏠 Post-build Actions

☒ Git

Repositories

Repository URL

https://github.com/manisha-a99/private-repo.git

Credentials

manisha-a99/*****

+ Add

Advanced

+ Add Repository

Branches to build

Save

Apply

 Jenkins

private-build

Configure

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Advanced

+ Add Repository

Branches to build

Branch Specifier (blank for 'any')

*/main

+ Add Branch

Repository browser

(Auto)

Save

Apply

Verify the status

 Jenkins

private-build

#1

private-build

Timings

Git Build Data

Cloning the remote Git repository

Cloning repository <https://github.com/manisha-a99/private-repo.git>

> git init /var/lib/jenkins/workspace/private-build # timeout=10

Fetching upstream changes from <https://github.com/manisha-a99/private-repo.git>

> git --version # timeout=10

> git --version # 'git version 2.50.1'

using GIT_ASKPASS to set credentials

> git fetch --tags --force --progress -- <https://github.com/manisha-a99/private-repo.git> +refs/heads/*:refs/remotes/origin/* # timeout=10

> git config remote.origin.url <https://github.com/manisha-a99/private-repo.git> # timeout=10

> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10

Avoid second fetch

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision acb5c926d2fb62ee888d7b3ed0559113d5f9c195 (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f acb5c926d2fb62ee888d7b3ed0559113d5f9c195 # timeout=10

Commit message: "Add 'hloo' to man file"

First time build. Skipping changelog.

Finished: SUCCESS